



Pharm*Achieve*

FEVER

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ASA Acetylsalicylic Acid

ED Emergency Department

GI Gastrointestinal

HCP Healthcare Provider

NSAIDs Nonsteroidal Anti-Inflammatory Drugs

PO By Mouth

PR Per Rectum

PRN As Needed

Q4H Every 4 Hours

Q6H Every 6 Hours

Q8H Every 8 Hours

PATHOPHYSIOLOGY AND CLINICAL PRESENTATION

Pathophysiology

- Fever is a regulated physiologic response to infections, malignancies, rheumatologic, and/or immunologic diseases
- Results in the body setting a new hypothalamic setpoint causing an elevation in core body temperature
- Fever is commonly defined as a rectal temperature $\geq 38^{\circ}\text{C}$
- Normal body temperature is known to fluctuate throughout the day by 0.5°C

Clinical Presentation

- General discomfort
 - Chills
 - Rigors
- At higher temperatures, malaise and fatigue is common but may not be attributed to the fever, rather, the underlying illness causing the fever

Fever alone is **not harmful** as it is the body's response to illness

GOALS OF THERAPY

- ✓ Increase patient comfort
- ✓ Provide symptomatic relief with treatment while minimizing adverse effects
- ✓ In pediatric patients, relieve parental anxiety
- ✓ In pregnant patients, reduce the risk of adverse fetal outcomes (especially in the first trimester)
- ✓ Assessing for, and when appropriate, addressing the underlying cause of fever

MEASUREMENT OF BODY TEMPERATURE

Method	Comments	Instructions For Use
Rectal (36.6-37.9°C)	• Gold standard for definitive measurement in children ≤5 years of age • Contraindicated in premature, immunocompromised, and rectal abnormalities	• Use cool and soapy water to clean the thermometer, then rinse • Adults: lay patient in fetal position, children: on their back with knees bent • Lubricate the thermometer and anus (pea-sized amount of petroleum jelly) • Insert the thermometer 2-3 cm into rectum with one hand, while holding it in place • Keep thermometer in place until it beeps and the temperature is shown
Oral (35.5-37.5°C)	• Recommended for confirmation of fever • First choice for children >5 years of age and in adults	• Avoid smoking, drinking, and eating 10 to 30 minutes prior to testing • Place thermometer under the tongue or between gum and cheek • Hold in place using lips or fingers (not teeth) and keep mouth closed • Keep thermometer in place until it beeps and the temperature is shown

MEASUREMENT OF BODY TEMPERATURE

Method	Comments	Instructions For Use
Axilla (36.5-37.5°C)	- Used in low-risk children <5 years of age - If fever is not shown, can confirm via another route - Disadvantages - length of time needed to get temperature reading and variations in readings - Advantages – accessible, safe, less frightening to children	- Place thermometer in centre of axilla - Keep arm tucked against the body to stabilize the thermometer - Hold the thermometer in place until it beeps and the temperature is shown
Tympanic (35.8-37.9°C)	- Can be used in children ≥2 years of age and in adults - May give low readings, not as reliable, can confirm via another route - Disadvantages – cost of thermometer - Advantages – speed, no external effects (i.e. beverages), no mucous membrane contact	- Follow manufacturers' directions carefully - Proper placement is very important to get an accurate reading (improper placement may give reading of outer ear canal wall) - Use a clean probe tip - Gently pull the ear back to straighten the ear canal - Insert thermometer into the ear until the canal is fully sealed off - Squeeze the button on the thermometer until it beeps and temperature is shown

RED FLAGS

- Age <6 months (ED if <3 months)
- Persistent fever for more than 3 days
- High fever >40.5°C
- Cranky, irritable, or fussy behaviour in children (or any other worrisome symptoms)
- Persistent coughing or wheezing
- Stiff neck, seizures
- Pain, swelling, or redness
- Confusion, lethargy, difficult to arouse
- New medication started recently
- Fever with a rash
- Recent travelling
- Recent cancer treatment

MEDICAL/DRUG RELATED CAUSES OF FEVER

Anticonvulsants

Idiosyncratic Reaction

Antimicrobials

Chemotherapy

Allopurinol

Drugs That Alter Thermoregulation

Immune Checkpoint Inhibitors

Parenteral Administration

Heparin

NONPHARMACOLOGICAL MEASURES

- Encourage adequate fluid intake
- Avoid physical exertion
- Remove excess clothing and bedding for comfort (use caution in infants/young children)
- Maintain environmental temperatures around 20-21°C
- *If* sponging is trialled due to a persistently high fever in addition to drug therapy, use lukewarm water (**do not use alcohol**)

PHARMACOLOGICAL MEASURES

To treat or not to treat:

- Fever is the body's normal physiologic response to inflammation/infection
- Fever is usually self-limiting and is rarely harmful
- Using antipyretics is generally **not required** but can be initiated to improve patient comfort
- The decision to treat a fever should always be **individualized**

PHARMACOLOGICAL MEASURES

- Reduces body temperature by reducing the thermoregulatory setpoint in the brain
- Intermittent administration is not recommended due to fluctuations in temperature and concomitant chills

Antipyretics: acetaminophen, ASA, ibuprofen, naproxen

Children

- Antipyretics used in children with a fever only if they appear distressed
- Antipyretics do not reduce the risk of febrile seizures (usually occurring in 6 months to 6 years of age)
- ASA is not recommended in children/adolescents due to risk of Reye's syndrome
- No clinical data available on the use of naproxen in children

Pregnancy

- Fever during the 1st trimester is associated with neural tube defects and multiple abnormalities
- Acetaminophen is considered 1st line
- ASA and NSAIDs should be avoided in the 1st and 3rd trimester (except low dose ASA where appropriate)

Breastfeeding

- Both acetaminophen and ibuprofen are considered safe

PHARMACOLOGICAL MEASURES

Drug Table

Drug	Dosage	Comments
Ibuprofen	- Children under 6 months: 5 mg/kg every 8 hours orally as needed; maximum 40 mg/kg/day; do not exceed the adult dose - Children over 6 months: 5–10 mg/kg every 6–8 hours by mouth as needed for symptom relief; maximum 40 mg/kg/day; do not exceed the adult dose - Adults: 200–400 mg every 4-6 hours by mouth as needed, maximum for self-care 1200 mg per day; maximum if supervised by health care provider is 2400 mg day	- AEs uncommon when used at the recommended dose for a short period of time - GI upset/pain - Avoid in renal dysfunction - Do not administer if dehydrated - Limited data available for the use in children <2 months of age
Naproxen	- Children under 12 years: not recommended - Adults: 220 mg every 8-12 hours orally as needed; maximum of 440 mg per day	- AEs uncommon when used at the recommended dose for a short period of time - Avoid in renal dysfunction - Do not administer if dehydrated

PHARMACOLOGICAL MEASURES (CONTINUED)

Drug Table

Drug	Dosage	Comments
ASA	- Children under 18 year; avoid use - Adults: 325–650 mg every 4-6 hours orally as needed; maximum of 4000 mg per day	- GI upset - Avoid in renal failure, peptic ulcer disease, heart failure and ASA-sensitive asthma
Acetaminophen	- Children: 10–15 mg/kg every 4–6 hours by mouth or rectally as needed for symptom management; maximum 75 mg/kg/day; do not surpass the adult dose - Adults: 325–650 mg every 4-6 hours by mouth or rectally as needed; maximum 4000 mg/day	- AEs uncommon when used at the recommended dose for a short period of time - Use caution in patients with active liver disease or dysfunction - Rectal administration should only be done under HCP supervision due to erratic absorption via this route

TAKEAWAY

Method	Normal Range	Notes
Oral	35.5-37.5°C	Commonly used; affected by eating, drinking, or smoking
Rectal	36.6-37.9°C	Gold standard for children ≤5 years of age
Axillary (Armpit)	36.5-37.5°C	Variable readings; less frightening for children
Tympanic (Ear)	35.8-37.9°C	Quick; technique-sensitive

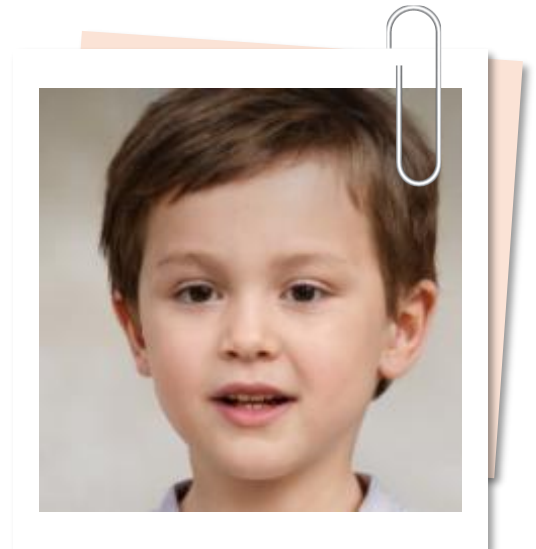
TAKEAWAY

Drug	Adult Dose	Pediatric Dose	Notes
Ibuprofen	200-400 mg Q4-6H PO PRN	<6 months: 5 mg/kg Q8H PO PRN > 6 months: 5-10 mg/kg Q6-8H PO PRN	• GI upset/pain • Avoid in renal dysfunction
Naproxen	220 mg Q8-12H PO PRN	<12 years old: Not recommended	Avoid in renal dysfunction
ASA	325-650 mg Q4-6H PO PRN	<18 years old: Not recommended	Avoid in children due to Reye's syndrome risk
Acetaminophen	325-650 mg Q4-6H PO/PR PRN	10-15 mg/kg Q4-6H PO/PR PRN	Use caution in patients with active liver disease or dysfunction

CASE

JC is a 4-year-old boy who was brought to your clinic by his mother because he has had a high fever for the past 2 days, peaking at 39.2°C (102.6°F). His mother reports that he has been more irritable than usual, has had a poor appetite, and has been sleeping more during the day. She also mentions he had a runny nose and a mild cough that started around the same time as the fever. On examination, JC is alert but clingy. He has no rash, no signs of ear infection, and no signs of dehydration. His mother is very concerned and asks if there is any medicine she can give him at home to bring the fever down.

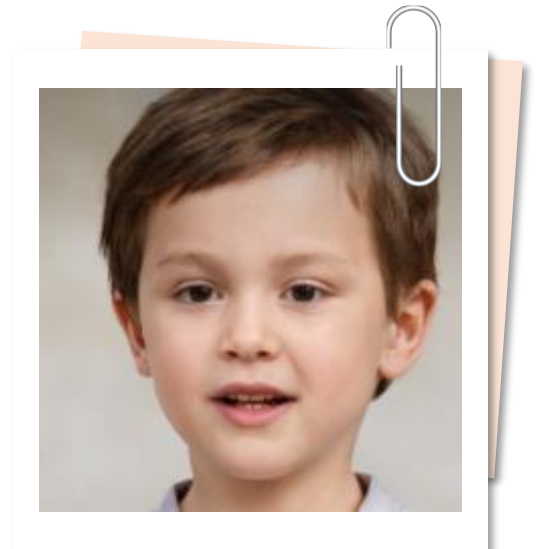
1. Which of the following is the MOST appropriate fever management recommendation for JC?
 - a) Aspirin® 10 mg/kg every 6 hours as needed
 - b) Acetaminophen 15 mg/kg every 4–6 hours as needed
 - c) Naproxen 250 mg twice daily
 - d) Ibuprofen 20 mg/kg every 6-8 hours as needed



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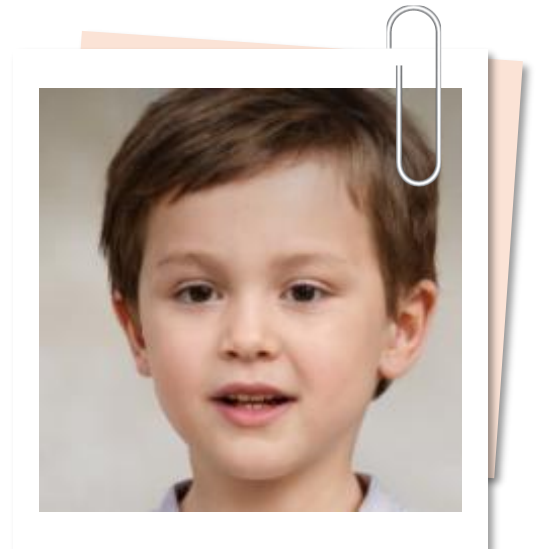
2. Which of the following findings would NOT be considered a red flag in JC?
- a) Fever duration of 24 hours
 - b) New-onset seizure during the fever episode
 - c) Cool, mottled extremities
 - d) Persistent vomiting and refusal to drink



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3. Which of the following is NOT an appropriate recommendation for JC's mother?
- a) Encourage the child to drink plenty of fluids
 - b) Use sponging with alcohol to cool the child
 - c) Maintain a room temperature of around 20 – 21 °C
 - d) Avoid physical exertion and remove excess clothing



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CHANGE LOG

April 2025

- Formatting changes throughout
- Slide 2: Updated legend
- Slide 5: Updated instructions for thermometer use
- Slide 6: Updated comments and instructions for thermometer use
- Slide 8: Updated red flags
- Slide 9: Updated medical/drug-related causes
- Slide 13: Updated drug table
- Slide 15: Added summary table
- Slide 16: Added summary table
- Slide 17, 18: Added case and case questions
- Slide 19: Updated references

June 2025

- Slide 18: Updated case question

October 2025

- Minor formatting changes made

March 2026

- Slide 2: Added NAPRA competencies
- Slide 4: Changed fever threshold to $\geq 38^{\circ}\text{C}$
- Slide 6: Updated guidance re oral measurement
- Slide 8: Simplified red flags
- Slide 12: Minor update to the use of ASA in pregnancy
- Slide 20: Updated references